

Model Meaning of `arcDefiningBodies`

Purpose

In a multi-arc estimation model, each estimated body's trajectory may be split into local arcs. The argument `arcDefiningBodies` specifies which body's local arc should define the state transition and sensitivity matrix used at a given epoch.

Multi-arc state model

Let body b have local arcs

$$I_{b,k} = [t_{b,k}^s, t_{b,k}^e],$$

with a separate estimated initial state for each arc,

$$x_{b,k,0} = x_b(t_{b,k}^s).$$

Inside that arc, the state is a function of the local initial state and global parameters p :

$$x_b(t) = \varphi_{b,k}(t; x_{b,k,0}, p), \quad t \in I_{b,k}.$$

The variational matrix for an active arc is, schematically,

$$M_k(t) = [\Phi_k(t) \quad S_k(t)] = \begin{bmatrix} \frac{\partial x_k(t)}{\partial x_{k,0}} & \frac{\partial x_k(t)}{\partial p} \end{bmatrix}.$$

For an observation model

$$h(t) = h(x_{b_1}(t_1), x_{b_2}(t_2), \dots, p),$$

the partial derivative with respect to estimated arc parameters uses the variational matrix of the arc that is considered active for the body involved:

$$\frac{\partial h}{\partial q} = \frac{\partial h}{\partial x_b(t)} M_{\alpha(t)}(t),$$

where $\alpha(t)$ is the selected arc index. The role of `arcDefiningBodies` is to define $\alpha(t)$.

Arc selection rule

Let D be the set passed through `arcDefiningBodies`.

$$\alpha_D(t) = \begin{cases} \text{default time-based arc,} & D = \emptyset, \\ \text{arc containing } t \text{ for the body or bodies in } D, & D \neq \emptyset. \end{cases}$$

If multiple bodies are provided, they must identify the same active arc. If one body says “arc 0” and another says “arc 1”, the model request is ambiguous: there is no single local variational matrix that represents both choices.

Example with distinct behavior

Consider two estimated bodies:

$$\begin{aligned} \text{Probe } P : \quad I_{P,0} &= [0, 100], \quad \text{arc index } 0, \\ \text{Relay } R : \quad I_{R,1} &= [20, 30], \quad \text{arc index } 1. \end{aligned}$$

The relay arc is completely inside the probe arc. At $t = 25$, both local models exist, but they correspond to different local initial states:

$$x_P(25) = \varphi_{P,0}(25; x_{P,0,0}, p), \quad x_R(25) = \varphi_{R,1}(25; x_{R,1,0}, p).$$

The selected variational matrix depends on D :

Input D	selected arc	returned matrix
$\{\}$	1	$M_1(25)$
$\{P\}$	0	$M_0(25)$
$\{R\}$	1	$M_1(25)$
$\{P, R\}$	conflict	no unique matrix

Thus the same epoch can produce different matrices:

$$\text{arcDefiningBodies} = \{P\} \quad \Rightarrow \quad \frac{\partial h}{\partial x_{P,0,0}} = \frac{\partial h}{\partial x_P(25)} M_0(25),$$

while

$$\text{arcDefiningBodies} = \{R\} \quad \Rightarrow \quad \frac{\partial h}{\partial x_{R,1,0}} = \frac{\partial h}{\partial x_R(25)} M_1(25).$$

At $t = 50$, only the probe arc is physically active:

$$50 \in I_{P,0}, \quad 50 \notin I_{R,1}.$$

Passing $D = \{P\}$ selects $M_0(50)$. Passing $D = \{R\}$ identifies no active relay arc. This is exactly the model-level reason for the input: an observation partial should use the local arc associated with the body whose state appears in that partial.

Interpretation

`arcDefiningBodies` is not a dynamical parameter. It is a rule for choosing the local variational solution in a multi-arc model. It matters when different estimated bodies have different arc intervals, overlapping arcs, or observations whose partials are associated with specific link-end bodies.